

## PROBLEM SET 4

### ECON301—ECONOMETRICS I

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**Note:** in the lecture notes and videos, I usually denote vectors by underlining them, for example  $\underline{x}$ , and I denote matrices in capital letters. When typesetting it, I use **bold notation and upright letters** to denote vectors and matrices, i.e.,  $\mathbf{x}$  and  $\mathbf{M}$  for example.

1. Consider the normal equations that determine the LSE:  $(\mathbf{X}^\top \mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$ . Let  $\mathbf{S}_{xx}$  be the  $k \times k$  in which the  $(i, j)$  element is  $S_{x_i, x_j} = \sum_{\ell=1}^n (X_{\ell i} - \bar{X}_i)(X_{\ell j} - \bar{X}_j)$ ,  $\mathbf{S}_{xy}$  be the  $k$  dimensional vector with elements given by  $S_{x_i, y} = \sum_{\ell=1}^n (X_{\ell i} - \bar{X}_i)(Y_{\ell} - \bar{Y})$ , and  $\hat{\boldsymbol{\beta}}_P = (\hat{\beta}_1, \dots, \hat{\beta}_k)^\top$  be the LSE estimator of the vector of slopes of the model. Show that  $\mathbf{S}_{xx}\hat{\boldsymbol{\beta}}_P = \mathbf{S}_{xy}$ .
2. Let  $\mathbf{X}$  be a regression matrix such that  $\mathbf{1} \in \mathcal{L}(\mathbf{X})$ . Show that  $\bar{Y} = \widehat{\bar{Y}}$  and  $\widehat{\bar{\epsilon}} = 0$ . *Hint: you can assume without loss of generality, that the vector of ones,  $\mathbf{1}$ , is part of the regression matrix  $\mathbf{X}$ .*
3. Let  $(X_i, Y_i), i = 1, 2, \dots$  be independent observations that fulfill assumptions 1M and 2M. Let  $\hat{\boldsymbol{\beta}}$  be the LSE estimator in a simple linear regression. Find and justify the limit in probability of  $\frac{1}{n} \sum_{i=1}^n Y_i^2$ ,  $\frac{1}{n} \sum_{i=1}^n \hat{\beta}_0 Y_i$ , and  $\frac{1}{n} \sum_{i=1}^n \hat{\beta}_1 Y_i X_i$ .
4. Finish the proof of Theorem 5.2.
5. The purpose of this exercise is to show that we always have

$$\text{SSE}(X_1, \dots, X_k, X_{k+1}) \leq \text{SSE}(X_1, \dots, X_k).$$

Consider

$$\text{SS}(c_0, c_1, \dots, c_k, c_{k+1}) = \frac{1}{n} \sum_{i=1}^n (y_i - c_0 - c_1 x_{i,1} - c_k x_{i,k} - c_{k+1} x_{i,k+1})^2,$$

and provide an argument using the idea of constrained optimization.